

birdclef-2026

13th place solution

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Thank you to the hosts of this competition, and to the participants who shared valuable insights throughout the competition. Many components of our solution use parts of public solutions and solutions from previous years, and this would not have been possible without them.

Overview

Our solution consists of 6 models converted with OpenVINO, with 3 CNNs and 3 SED models, ensembled at a 0.5-0.5 ratio with the 0.949 public scoring [raunak](#) version of the public Perch / ProtoSSM pipeline. On public LB, our 6 model ensemble scored 0.943 + perch/protossm pulled us to 0.957 on public. Our CNN/SED ensemble was better on private however - scoring 0.953. With perch, this increases to 0.955 (but at a 0.6-0.4 ratio ours-perch it scores 0.957 )

Dataset

2026 BirdCLEF train audio and (pseudolabeled) soundscapes.

Training

Sampling

Our model trains on random 5 second segments of audio. RMS sampling / sampling from the start did not perform well. Longer segments do not perform as well, but I kept a 10s model in the ensemble for diversity

Model

Our 3 CNNs are all trained using [tawara's](#) hgnet training notebook, with minimal changes. Our 3 SEDs are trained using [tucker arrant's sed](#) notebook, with more moderate changes including switching out the head, disabling perch distillation (it did not work for us when including pseudos), and changing the augmentations.

Loss

For both CNNs and SEDs, we use focal BCE loss. I tried CE loss, ASL loss, and AUC loss, and all performed poorly.

Data augmentation

For our CNNs, we only used mixup on mel spectrograms. For the effnetv2s specifically, adding the 2023 2nd place augmentations improved score. For our SEDs, we used mixup on mel spectrograms + the augmentations from the 2023 2nd place solution: One of (white noise, pink noise, gaussian noise) at $p = 0.3$, and gain at $p = 0.5$.

Mel spectrogram parameters

All models in each CNN / SED family respectively used the same mel spec parameters.

For CNNs: n_fft=2048, hop_length=313, win_length=626, n_mels=256, fmin=20, fmax=16000, power=2.0
For SEDs: n_fft=2048, hop_length=512, win_length=2048, n_mels=128, fmin=20, fmax=16000, power=2.0

Backbone

Our 3 CNN backbones are as follows - 1 hgnetv2b0, 1 effnetb3ns, 1 effnetv2s (2025 pretrained checkpoint from volodymyr)

Our 3 SED backbones are as follows - 2 effnetv2s (one using 10s windows), 1 eca-nfnet, all using 2025 pretrained checkpoints from volodymyr

Including pseudos, these models' scores range from 0.934-0.938. The 2025 pretrained checkpoints provided a significant boost (~0.007) on LB. Without pseudos, the models' scores are around 0.88-0.90, but I did not individually submit each of them without pseudos.

Pseudolabeling

I actually didn't perform an initial iteration of pseudolabeling. Instead, I used the publicly available pseudolabels generated by Ali Memetoglu [here](#). These pseudolabels were very strong and that alone pushed tawara's single model base hgnet that I was using from 0.881 to 0.934. I used the same pseudolabels to train all my models, and concatenated all the pseudolabels to the train audio for training.

Pseudolabel sampling in specific ratios did not work.

A second iteration of training on generated pseudolabels using the 6 model ensemble did not work. Combining perch logits on the pseudolabels and using that to train did not work either.

Other details

- 30 epochs
- 5e-4 LR
- One cycle scheduler

Inference and TTA

We use the 5s windows from soundscapes and feed into the models. For the 10s model, we use Nikita's 2025 strategy: sliding a 10s window with a 5s step across the 60s file, and padding each side by 2.5s so each window is centered on a 5s row. We also apply a 2.5 second shift for TTA on 5s models.

Ensemble and postprocessing

We ensembled all 6 models with equal weight, using a simple weighted average. We did not use any postprocessing as the public Perch pipeline already performed a lot of it and stacking it on our CNNs/SEDs performed poorly.

What did not work

- background noise
- pretraining on Xeno Canto myself
- perch distillation
- raw signal model

along with some other strategies already mentioned above